Problem	Penalty	Free	Informed	Free − Penalty	Informed − Free	COLSHADE	L-SHADE Deb	L-SHADE ε-long
lens/original	0.611 ± 0.015	**0.692 ± 0.008**	0.641 ± 0.037	+0.081*	−0.052	0.582	0.586	0.625
lens/realistic_draft	0.662 ± 0.023	**0.675 ± 0.021**	0.647 ± 0.013	+0.014	−0.028	0.579	0.582	0.620
cec/RC12	0.516 ± 0.005	0.718 ± 0.049	**0.732 ± 0.056**	+0.202*	+0.014	0.643	0.621	0.583
cec/RC13	**0.963 ± 0.017**	0.955 ± 0.005	0.953 ± 0.007	−0.008	−0.002	0.849	0.795	0.758
cec/RC15	0.940 ± 0.031	**0.960 ± 0.010**	0.953 ± 0.006	+0.020	−0.006	0.711	0.696	0.680
cec/RC16	**0.563 ± 0.036**	0.561 ± 0.055	0.516 ± 0.023	−0.002	−0.044	0.486	0.496	0.493
cec/RC18	**0.648 ± 0.041**	0.622 ± 0.015	0.635 ± 0.033	−0.026	+0.013	0.591	0.606	0.622
cec/RC21	**0.935 ± 0.036**	0.885 ± 0.049	0.926 ± 0.018	−0.050	+0.041	0.811	0.743	0.729
cec/RC25	0.511 ± 0.004	**0.535 ± 0.014**	0.529 ± 0.009	+0.024*	−0.006	0.515	0.514	0.510
cec/RC26	**0.557 ± 0.017**	0.518 ± 0.012	0.513 ± 0.009	−0.038*	−0.006	0.375	0.427	0.418
Mean	0.691 ± 0.008	**0.712 ± 0.003**	0.705 ± 0.008	+0.022*	−0.008	0.614	0.607	0.604

Test AOCC (higher is better). LLaMEA: mean ± std over 5 runs; baselines: mean over 15 seeds.

*Bold: best in row. * $p < 0.05$ (Mann–Whitney U, uncorrected).*